Two-Dimensional Defect-Parameter Validation of the SolarLab Drift–Diffusion Simulator against SCAPS-1D at the Perovskite/ETL Interface

23 June 2026

Abstract

The SolarLab one-dimensional drift–diffusion solar-cell simulator is benchmarked against the established reference solver SCAPS-1D on two two-dimensional defect-parameter scans of the perovskite/electron-transport-layer (ETL) interface: interface defect density versus trap energy ($N_t \times E_t$), and interface defect density versus conduction-band offset ($N_t \times \Delta E_C$). Both maps are reproduced on SolarLab's validated parity configuration using the transient current–voltage driver. SolarLab recovers the qualitative structure of all four figures of merit across both scans, including the defect-density banding, the shallow- versus deep-trap response, and the coupling between the conduction-band cliff/spike and the interface defect density that the second scan was designed to expose. The residual quantitative differences are confined to two previously documented model boundaries: a bulk-Auger-limited open-circuit-voltage ceiling (1.169~V versus SCAPS 1.250~V at low defect density) and the spike-side collapse mechanism (fill-factor and short-circuit-current loss in SolarLab versus open-circuit-voltage collapse in SCAPS). No new discrepancies are observed, and SolarLab converges on every grid point whereas SCAPS leaves approximately twelve points unresolved.

1. Introduction

A device simulator earns confidence by reproducing the *direction and shape* of a cell's response to its design parameters, not merely by matching a single operating point. Single-variable sweeps validate one parameter at a time; two-dimensional scans additionally test whether the simulator captures the *coupling* between two parameters. The reference SCAPS-1D study concludes with two such scans on the perovskite/ETL interface, chosen because that interface was the most sensitive location in the preceding single-variable analysis. This report asks whether SolarLab can perform the same two scans and how its maps compare with the SCAPS reference.

2. Methodology

Each scan is a nested two-parameter sweep (it is not the two-dimensional spatial solver). SolarLab's parameter-sweep interface applies both axis values in a single update, targeting the perovskite/ETL interface defect. All runs use the validated parity configuration `scaps_mirror_v2`, which combines the effective density-of-states band-potential correction with the heterointerface

Auger de-spike ($f = 0.53$), and the transient (method-of-lines) current–voltage driver. The transient driver is used in preference to the steady-state Newton driver because it remains convergent through the collapsed-junction cliff and spike regimes, reproducing SCAPS's steady-state-per-bias result as the fast-scan, ion-frozen limit.

The two scans follow the SCAPS definitions exactly:

- **Scan A** ($N_t \times E_t$): interface defect density $N_t = 10^9$ to 10^{15} cm^{-2} (seven levels) by trap energy $E_t = 0.01$ to 0.6 eV (eight levels), giving $7 \times 8 = 56$ points.
- **Scan B** ($N_t \times \Delta E_C$): the same seven defect-density levels by the ETL/PVK conduction-band offset $\Delta E_C = -1.0$ to $+0.70$ eV (sixteen levels), giving $7 \times 16 = 112$ points. The offset is imposed through the electron affinity, $\chi_{\text{PVK}} = 3.94$ eV fixed and $\chi_{\text{ETL}} = 3.94 - \Delta E_C$.

At every grid point the four figures of merit (PCE, V_{oc}, FF, J_{sc}) are extracted from the simulated current–voltage curve; points without an open-circuit crossing are recorded as undefined, mirroring the unresolved cells in the SCAPS maps.

For legibility, the SCAPS reference maps are colour-digitised and re-rendered at the same scale and font size as the SolarLab maps; the digitisation was validated against the reported best-point values and reproduces them to the displayed precision (e.g. Scan~A: PCE~$= 29.83\,\%$, $V_{oc} = 1.250$~V).

3. Results

3.1 Interface defect density versus trap energy

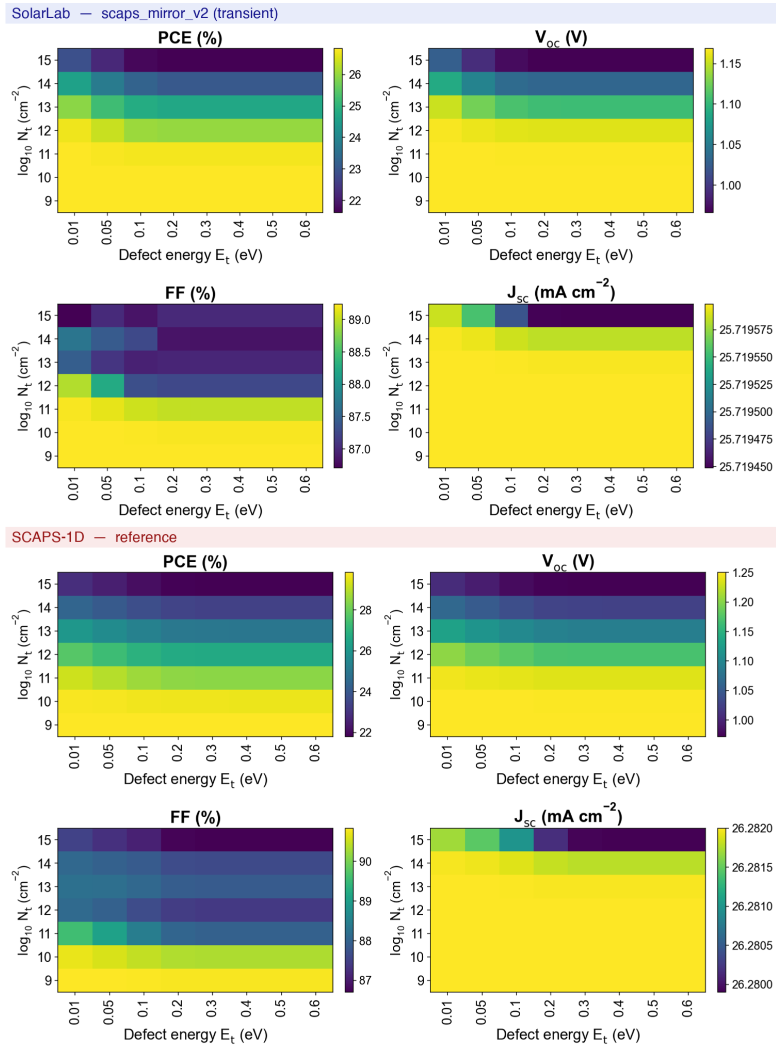

Figure 1: Scan A ($N_t \times E_t$) on the perovskite/ETL interface. SolarLab (upper block) and SCAPS-1D (lower block) four-panel maps of PCE, V_{oc}, FF and J_{sc}; the vertical axis is $\log_{10} N_t$ and the horizontal axis is the trap energy E_t.

Both maps are banded predominantly by defect density: low N_t yields high V_{oc} and PCE, and the cell collapses as N_t increases. Superimposed on this banding is a weaker trap-energy modulation in which a shallow trap ($E_t = 0.01$ eV) is least harmful and deeper traps reduce V_{oc} before saturating beyond $E_t \approx 0.2$ eV. The short-circuit current is essentially flat in both solvers. The single systematic difference is the open-circuit-voltage ceiling at low defect density: SolarLab saturates at 1.169 V whereas SCAPS continues to 1.250 V, because once interface recombination is removed SolarLab's V_{oc} is limited by bulk Auger recombination.

3.2 Interface defect density versus conduction-band offset

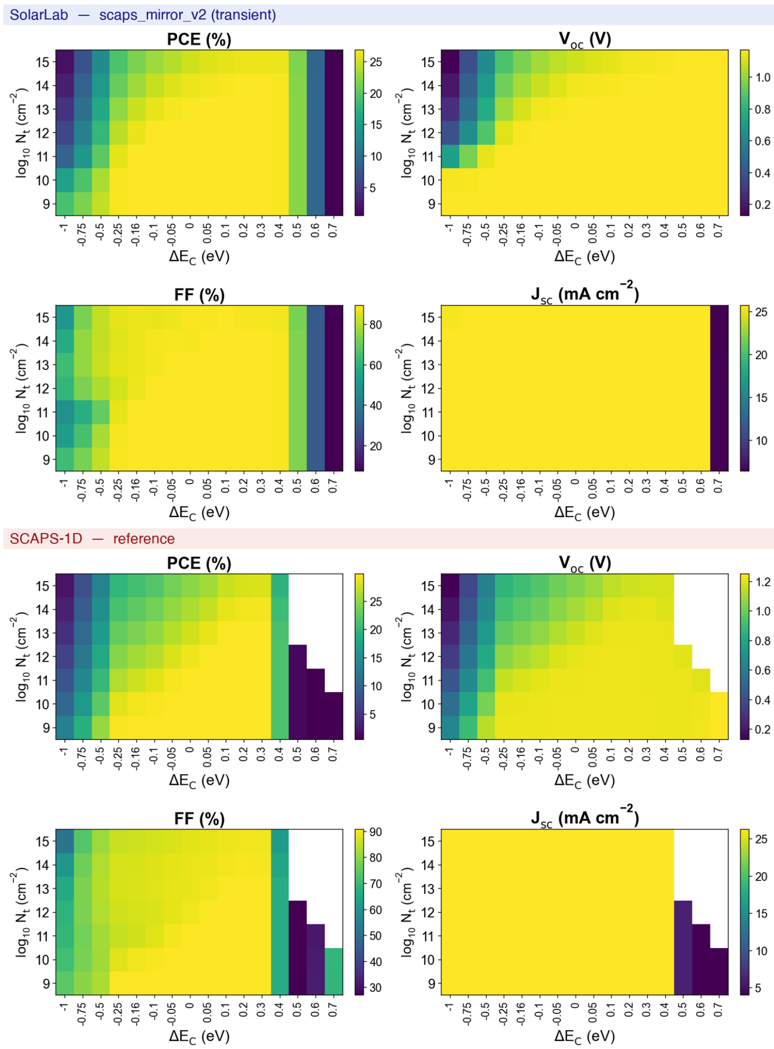

Figure 2: Scan B ($N_t \times \Delta E_C$) on the perovskite/ETL interface. Panels and axes as in Figure 1, with the horizontal axis the conduction-band offset ΔE_C. White cells in the SCAPS block are grid points the reference solver did not resolve.

Both maps display a high-performance plateau for $\Delta E_C \approx -0.5$ to $+0.4$ eV, a defect-density-dependent *cliff* triangle for $\Delta E_C \leq -0.5$ eV, and a *spike* collapse for $\Delta E_C \geq +0.5$ eV. SolarLab reproduces the cliff–defect coupling quantitatively: at $\Delta E_C = -1.0$ eV the open-circuit voltage barely falls at low defect density (1.165 V at $N_t = 10^9$) but collapses at high defect density (0.129 V at $N_t = 10^{15}$), confirming that the band-offset cliff is detrimental only when interface defects are present.

Two differences are noted. First, the spike-side collapse proceeds by a different channel: SCAPS collapses the open-circuit voltage (to about 0.2 V) and the solver fails on approximately twelve grid points (white cells), whereas SolarLab holds $V_{oc} \approx 1.169$ V and instead collapses the fill factor (from 89 to 30 $\%$ at $\Delta E_C = +0.6$ eV) and then the short-circuit current (to 6.4 mA cm^{-2}

at $\Delta E_C = +0.7$ eV), an S-shaped collection failure rather than a voltage collapse. Second, SolarLab's transient driver converges on all 112 points, where SCAPS leaves approximately twelve unresolved.

4. Discussion

The two scans confirm that SolarLab captures the physics structure of the two-dimensional defect/band-offset landscape on every axis the scans were designed to interrogate: the defect-density banding, the shallow- versus deep-trap response, and – most significantly – the coupling between the conduction-band cliff/spike and the interface defect density. The quantitative residuals coincide exactly with the two boundaries already identified in the single-variable parity campaign. The open-circuit-voltage ceiling reflects a bulk-Auger recombination floor that limits SolarLab's voltage once interface recombination is suppressed; it is a known absolute-magnitude offset, not a structural disagreement. The spike-side collapse channel reflects the documented difference in how the two solvers treat the strongly band-offset junction; SolarLab's robustness there (no unresolved points) is a property of its transient time-integration.

Table 1 summarises the base operating points and Table 2 the qualitative agreement per behaviour.

Table 1: Best-performance operating point of each scan (lowest defect density).

Metric	SolarLab A	SCAPS A	SolarLab B	SCAPS B
PCE (%)	26.8	29.83	26.9	29.86
V_{oc} (V)	1.169	1.2495	1.169	1.2511
FF (%)	89.2	90.84	89.3	90.82
J_{sc} (mA cm^{-2})	25.7	26.28	25.7	26.28

Table 2: Qualitative agreement between SolarLab and SCAPS-1D.

Behaviour	Agreement	Note
N_t banding (both scans)	match	—
Shallow/deep E_t response	match	—
Cliff–defect coupling	match	V_{oc} $1.165 \to 0.129$ V across N_t
V_{oc} ceiling at low N_t	differ	bulk-Auger floor (1.169 vs 1.25 V)
Spike collapse channel	differ	FF/J_{sc} vs V_{oc}
Unresolved grid points	—	$0/112$ (SolarLab) vs ~ 12 (SCAPS)

5. Conclusion

SolarLab reproduces both SCAPS-1D two-dimensional interface-defect scans and agrees with the reference on every qualitative trend they were constructed to probe. The two residual differences – the bulk-Auger open-circuit-voltage ceiling and the spike-side collapse mechanism – are the same documented model boundaries established in the single-variable validation, not new discrepancies. The comparison therefore extends the trends-over-absolutes validation of SolarLab from one-dimensional sweeps to the coupled two-parameter regime. All results, heatmaps and the generating script are archived under `outputs/scan2d/`.